DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA
Politecnico di Milano

Machine Learning (Code: 097683)
June 24, 2024

Surname:	
Name:	
Student ID:	
Row:	Column:

Time: 2 hours — Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 30 min before the end of the exam you will get 3 points, if you finish earlier than 20 min you will get 2 points and if you finish earlier than 10 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Discuss at least three methods used for feature selection and compare their advantages and disadvantages.

Exercise 2 (7 marks)

Explain the concept of bagging (Bootstrap Aggregating) in machine learning. Describe a scenario where bagging would be particularly beneficial.

Exercise 3 (2 marks)

Consider the following snippet of code:

```
1  X = dataset[['sepal-length', 'sepal-width', 'petal-length']].values
2  t = dataset['petal-width'].values
3
4  N, M = X.shape
5  lambda_ = 0.1
6
7  Phi = np.hstack((np.ones((N,1)), zscore(X)))
8  w1 = inv(Phi.T @ Phi) @ (Phi.T @ t)
9  w2 = inv(Phi.T @ Phi - lambda_ * np.eye(M + 1)) @ (Phi.T @ t)
10 w3 = (np.abs(w1) > 1/lambda_) * 1/lambda_ + (np.abs(w1) <= 1/lambda_) * w1
```

1. Describe what the code is doing from lines 1 to 9. What kind of machine learning problem is it attempting to solve? What machine learning techniques are being used?
2. Tell if lines 7-9 are correct. If not, identify the mistakes and propose a correction.
3. What happens when increasing the value of `lambda_` to `w2` and `w3`? Describe the role of `lambda_` in terms of bias-variance trade-off.

1. From line 1 to line 9, the code is loading the Iris dataset using three variables as input and the fourth one as target. Then, it constructed a design matrix `Phi` including the intercept term and performing the normalization with `zscore` for the other variables. Then, OLS is applied to compute `w1` while Ridge regression is used for computing `w2`, with regularization parameter `lambda_`. The code is attempting to solve a regression problem. The techniques are OLS (ordinary least squared), i.e., linear regression without regularization and linear regression with Ridge regularization.
2. Line 9 is not correct because of the minus sign in the application of the regularization. It should be replaced with `w2 = inv(Phi.T @ Phi + lambda_ * np.eye(M + 1)) @ (Phi.T @ t)`.
3. When increasing the value of `lambda_` we are enforcing more regularization. This is straightforward for Ridge regression when computing `w2`. It can be seen that for `w3`, instead, `1/lambda_` defined a clipping threshold, so that if the weight obtained by OLS is above `1/lambda_` in absolute value, it is clipped to `1/lambda_`. Thus, the threshold approaches zero as `lambda_` increases. In term of bias-variance trade-off, increasing `lambda_` reduces the variance but increases the bias of the model.

Exercise 4 (2 marks)

Tell if the following statements about Markov Decision Processes (MDPs) are true or false. Motivate your answers.

1. If the MDP has stochastic rewards, it may not admit a deterministic optimal policy.
2. $V^*(s) \geq Q^*(s, a)$ for all states s and actions a, where V^* is the optimal value function and Q^* is the optimal state-action value function.
3. A larger discount factor assigns more importance to the long-term effects of actions.
4. Value iteration is guaranteed to converge to the optimal value function in a finite amount of steps.

1. False. Every MDP admits a deterministic optimal policy. Stochastic rewards do not matter, since the optimal policy can just be defined in terms of the *expected* rewards of state-action pairs.
2. True, since $V^*(s) = \max_a Q^*(s, a)$.
3. True, provided some actions do have long-term effects, since a larger discount factor gives more weight to rewards further in the future.
4. False, value iteration is only guaranteed to converge to the optimal value function asymptotically.

Exercise 5 (2 marks)

Consider the parametric classification model $f_{\boldsymbol{w}}(x) = \text{sign}\left(\sum_{i=0}^{d} w_i x^i\right)$, where $\boldsymbol{w} = [w_0, \ldots, w_d]^\top$ and $d \in \mathbb{N}$ that is trained on a dataset $\mathcal{D} = \{(x_i, t_i)\}_{i=1}^{N}$ made of N samples. Tell if the following statements are true or false. Motivate your answers.

1. Keeping N fixed and increasing d, the variance of the model reduces.
2. Keeping d fixed and increasing N, the bias of the model reduces.
3. Increasing both N and d, the variance of model increases.
4. Decreasing both N and d, the bias of model increases.

1. FALSE: By increasing the degree d of the polynomial, we are considering enlarging hypothesis spaces, and, therefore, the variance does not reduce.
2. FALSE: By increasing the number of samples N, we are not affecting the bias that remains fixed since the degree d does not change.
3. FALSE: By increasing the number of samples N with a fixed degree d, we reduce the variance, but, by increasing d with a fixed N, we increase the variance. Therefore, we can conclude nothing about the variance, in this case.
4. TRUE: The number of samples N does not affect the bias, but by reducing the degree d, we are considering simpler models and, therefore, we are increasing the bias.

Exercise 6 (2 marks)

Consider the following real-world problem. An online bookstore wants to recommend books to its potential clients. Whenever a user enters the homepage of the website, they are immediately shown a single book from the bookstore's catalog. We want to display the book that maximizes the probability that the user will end up buying it. For each of the following situations, briefly discuss how the problem could be modeled and solved using the techniques seen in the class:

1. The bookstore does not have any previous purchase data, nor any information about the current visitor.

2. The bookstore has access to detailed information (such as age, location...) and previous purchases of registered users.

1. This is a hard problem, but we can still model it as a multi-armed bandit where the arms are the books, and try to recommend to all users the single book that has the largest probability of being bought overall. We could solve this with UCB or Thompson sampling.

2. The purchase history of each registered user can be seen as a Markov decision process. The state can be designed to summarize the user's data and purchase history, and the actions are the books to recommend at each visit of the same user. To solve the problem, we can use reinforcement learning control methods such as SARSA or Q-learning, deploying a separate instance for each registered user. If we already have a large amount of purchase data, we could also use an off-policy (offline) RL approach. Alternatively, we can also use multi-class classification with features based on the user's data and history, and the classes being the books from the catalog.

Exercise 7 (4 marks)

Consider a binary classifier, defined in terms of the threshold $\iota \in [0,1]$:

$$y_{\boldsymbol{w}}(\boldsymbol{x}) = \begin{cases} 1 & \text{if } \sigma(\boldsymbol{w}^\top \boldsymbol{x}) \geq \iota \\ 0 & \text{otherwise} \end{cases}, \tag{1}$$

where $\sigma(\cdot)$ is the sigmoid function. The weights $\boldsymbol{w}$ are trained with Logistic Regression on a training set made of $N = 100$ samples and the threshold is set to $\iota = 1/2$.

1. Supposes that Accuracy = 0.4, Recall = 0.5, and that the positive samples correctly classified are 20, draw the confusion matrix.
2. Compute the F1 score.
3. Suppose we move the threshold ι to the value $1/4$, and we discover that the number of samples wrongly classified as positive does not change. What can we conclude about the Precision? Provide a formal justification.

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1. Knowing that Accuracy = 0.4 and Recall = 0.5, we deduce:

$$\text{Accuracy} = \frac{TP + TN}{N} = 0.4 \implies TP + TN = 40, \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} = 0.5 \implies TP = FN. \tag{3}$$

Thus, exploiting that $TP = 20$, we have the system:

$$\begin{cases} TP + TN = 40 \\ TP = FN \\ TP = 20 \end{cases} \implies \begin{cases} TP = 20 \\ TN = 20 \\ FN = 20 \end{cases}. \tag{4}$$

From which, $FP = N - TP - TN - FN = 40$. Thus, the confusion matrix becomes:

	Actual Class: 1	Actual Class: 0
Predicted Class: 1	20	40
Predicted Class: 2	20	20

2. We first compute the Precision:

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{20}{20 + 40} = \frac{1}{3}. \tag{5}$$

The F1 score is the harmonic mean between Precision and Recall:

$$F = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} = \frac{2}{3 + 2} = \frac{2}{5}. \tag{6}$$

3. By decreasing the threshold to $\iota = 1/4$, there will be more samples classified as positive. In particular, $TP' \geq TP$. Since the text says that $FP' = FP$, we have:

$$\text{Precision}' = \frac{TP'}{TP' + FP'} = \frac{1}{1 + \frac{FP}{TP'}} \geq \frac{1}{1 + \frac{FP}{TP}} = \frac{TP}{TP + FP} = \text{Precision}. \tag{7}$$

Thus, the precision increases.

Exercise 8 (4 marks)

Consider a linear binary SVM classifier that, after training, has weights $w = [-1, 2]$ and bias $b = -3$. Motivate all answers and write down any necessary computation.

1. How is the point $[0.5, 1]$ classified?
2. Provide an example of a support vector.
3. Suppose the point [1,1] is added to the dataset, with a negative label. Do we need to retrain the SVM?
4. Can we discard the point $[-3, 2]$ from the dataset if all we care about is predicting the class of new points?

1. $f([0.5, -1]) = -1 * 0.5 + 2 * 1 - 3 = -1.5 < 0$, so the point is classified as negative.
2. An example is $[0, 1.5]$, since $f([0, 1.5]) = -1 * 0 + 2 * 1.5 - 3 = 0$, so it is on the boundary. Any vector v with $|f(v)| \leq 1$ is a good example.
3. No, since $f([1, 1]) = -1 * 1 + 2 * 1 - 3 = -2 < -1$: the point is correctly classified and is not a support vector, so the classifier would not change.
4. With a linear SVM, we can discard the whole dataset, the weights are enough! If we are using a non-parametric version (e.g. a kernel-based implementation with a linear kernel), we have to check if the point is a support vector. $f([-3, 2]) = -1 * (-3) + 2 * 2 - 3 = 4 > 1$, so it is not a support vector and it can be discarded anyway.